

COURSE TITLE

DRILLING ENGINEERING I

FALL 2014

Dr. Samuel O. Osisanya, P. E.
Professor
Petroleum Engineering Department
The Petroleum Institute
Abu Dhabi
United Arab Emirates

Drilling Geology

Objectives of this lecture:

- Name the three classifications of rock.
- State what is meant by a Clastic rock and list the processes involved in forming Clastic rocks.
- Name the three types of sedimentary rock, based on how they were formed.
- Name drilling problems associated with several rock types

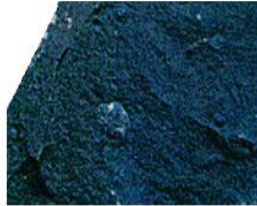
3 Classifications of Rocks

Igneous rock
Granite



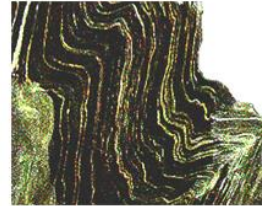
Molten rock
cooling and
crystallizing

Sedimentary rock
Shale



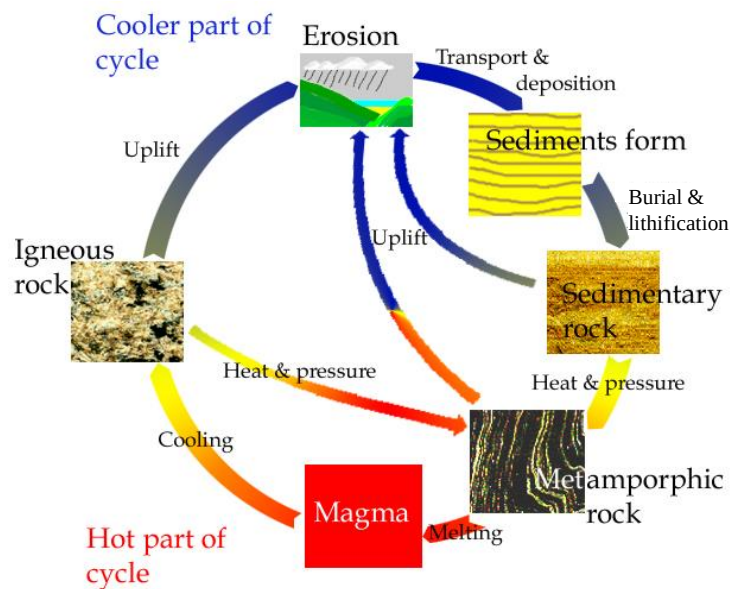
Rock fragments
becoming joined
together

Metamorphic rock
Gneiss



Existing rock
changing with high
temp / pressure

The Life Cycle of Rock



Sedimentary Rocks

Sedimentary rocks have three sources.

- Clastic (Fragmental). Forms from rock particles and skeletal remains. Example; sandstone, shale.
- Chemical. Deposited by the evaporation of water. Example; salts, gypsum.
- Biological. Remains of plant and animal life. Example; coal, coral.

Clastic Rock Formation

There are six processes for clastic sedimentary rock formation.

- Erosion. Existing rocks are broken down.
- Transport. Wind or water move the rock fragments.
- Deposition. Rock fragments are laid down in beds.
- Compaction. Burial decreases the sediment volume.
- Cementation. Minerals grow in the spaces between grains.
- Diagenesis. Chemical changes to fragments to form rock.

Engineers Training for Drilling Can Cheer Daily!

Sedimentary Rocks Relative Abundance



Shales - dominant sedimentary rock

Sandstone and conglomerates

Limestone and dolomite

- Shale are the dominant sedimentary rock comprising 75% of the total.
- Sandstones and conglomerates account for 11% of the total worldwide.
- Limestones and dolomites comprise 13%
- All others together make up about 1% (evaporites, coals)

Drilling Shales

- Reactive shale minerals hydrate and destabilize
- Fractured or stressed shales can be very unstable
- Competent shales are suitable for setting casing
- Shales cause 90% of all geological hole problems!

Drilling Limestone

- Fractured and vugular limestones may cause total losses and stuck pipe
- Bentonite based mud systems may **flocculate** with drilled limestone
- Chert inclusions can destroy most bits
- Unfractured limestone can make a good casing point

Drilling Coal

Coal is brittle and is often fractured. If not already naturally fractured, drilling stress may fracture it.

- Stuck pipe due to blocks of coal falling in
- Gas may be present
- Overgauge hole and ledges
- Total losses in fractured coal
- Can ignite if drilled with air

Drilling Salts

Under pressure and temperature salt is a plastic material and will flow or extrude into the hole

- Very high pressure kicks possible in or just below the salt
- Muds and cement slurries must be designed for the particular salts present
- Casing must be designed to withstand mobile formation
- Hole size may close behind the bit or bottom hole assembly preventing pulling the drillstring, may cause torque and drag and ultimately lead to stuck pipe

Drilling Geology 2

Crustal Plates

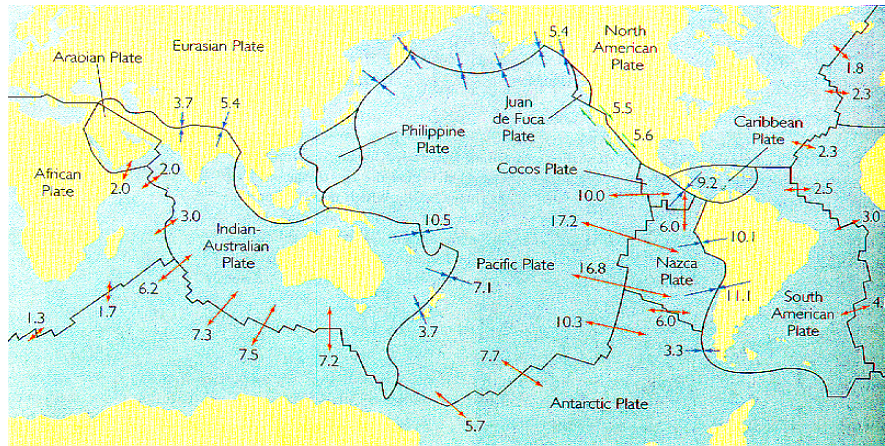
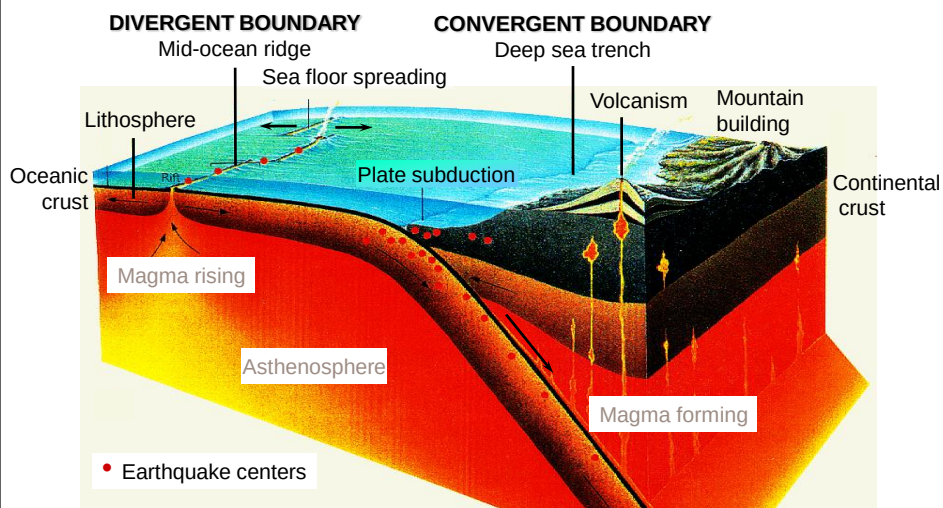


Plate boundaries shown with relative velocities (cm/yr)

Basic Elements of Plate Tectonics



Processes at Plate Boundaries

Divergent boundary - plates moving apart.
Earthquakes, new crust formed.

Convergent boundary - plates moving together.
Earthquakes, volcanoes, mountain formation.

Sliding boundary - plates sliding past one another.
Earthquakes.

All of these cause abnormal rock stresses

Rock Deformation From Stress

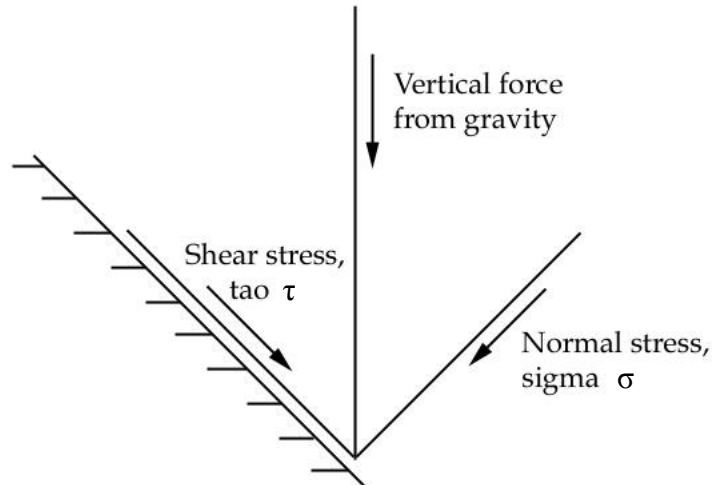
Stress in rock causes various deformations;

- Folding
- Faulting
- Fracturing
- Changes in porosity (compact/expand)

Plate movement, gravity and heat flow provide energy for the large scale processes (plate tectonics).

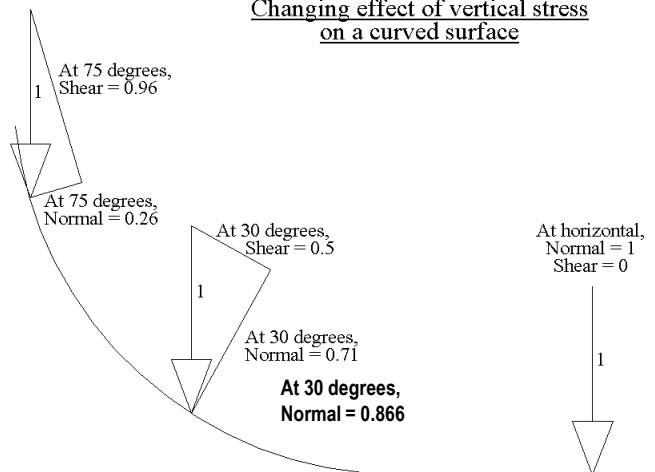
Salt diapirs and fluid movements cause smaller scale stresses.

Resolving Vertical Stress Into Vectors



Normal and Shear Stresses

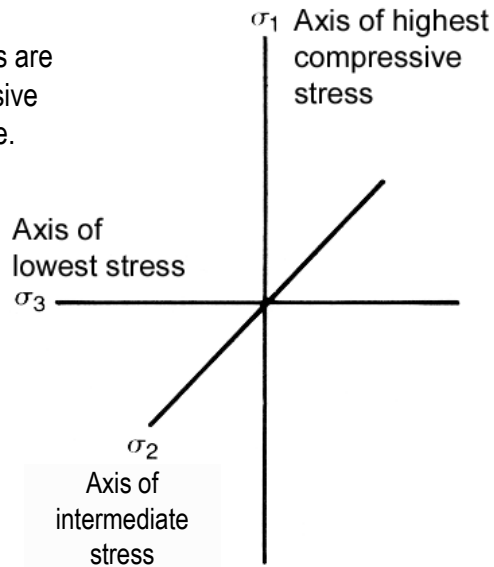
Changing effect of vertical stress
on a curved surface



The vertical stress = 1 psi/ft

Principal Stresses in Normal Rock

Principal stresses are usually compressive but *can be* tensile.



Highest Principal Stress, σ_1

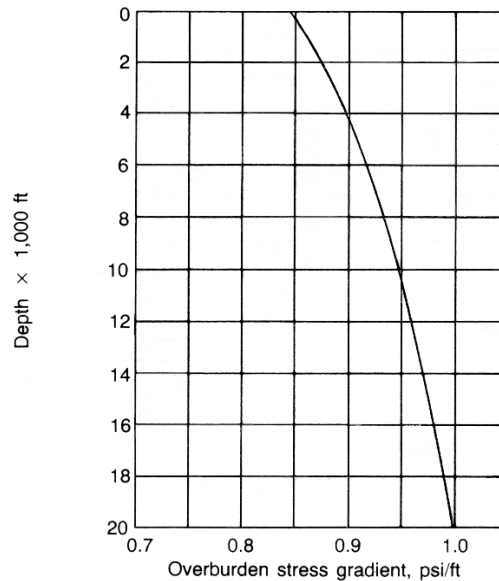
σ_1 = stress due to overburden (normal area).

The overburden is measured by integrating a Density Log to calculate the total weight of the rocks, which is divided by TVD.

σ_1 *might not be vertical in;*

- tectonically active areas
- close to salt diapirs
- if very deep
- close to faults

σ_1 Vs. Depth (US Gulf Coast)



(Preston Moore,
Drilling Practices
Manual, 2nd Ed,
Pennwell)

Least Principal Stress, σ_3

σ_3 is normally horizontal.

The **Eaton Equation** can be used to calculate σ_3 if Poisson's Ratio is known.

A Leakoff Test will give a direct indication of σ_3 (equals fracture closure pressure)

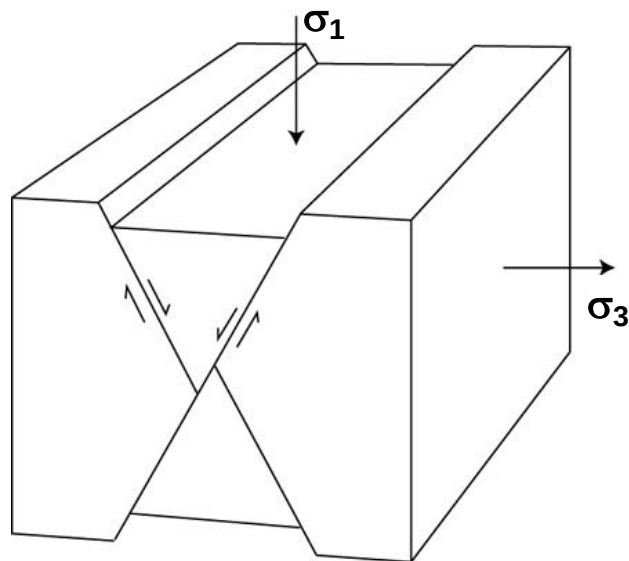
Intermediate Principal Stress, σ_2

σ_2 is normally horizontal.

$\sigma_2 \approx 1.1 \times \sigma_3$ (valid for the North Sea). This is only true in tectonically relaxed areas

Horizontal stresses can be estimated from offset well instability (4 arm caliper log) or regional trends.

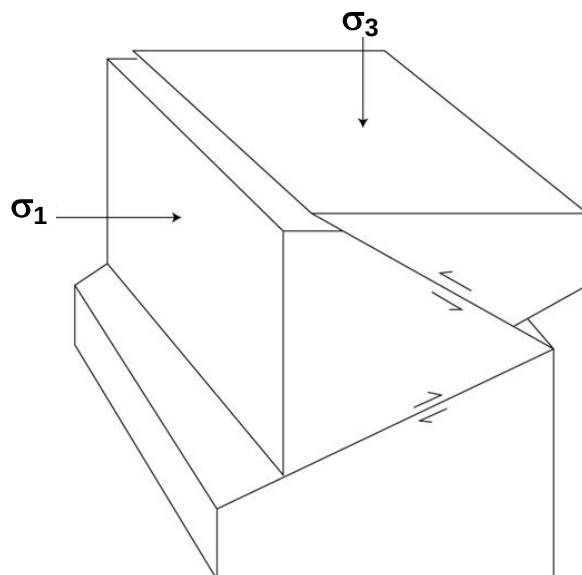
Normal Fault (Crust Stretches)



Normal Fault



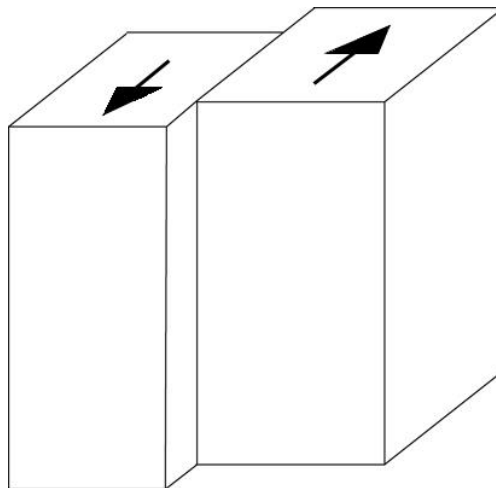
Thrust Fault (Crust Compresses) – Also called Reverse Fault



Thrust Fault

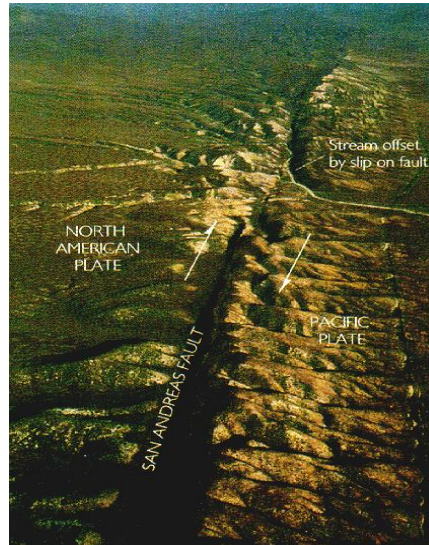


Strike-slip Fault (Crust Slides)

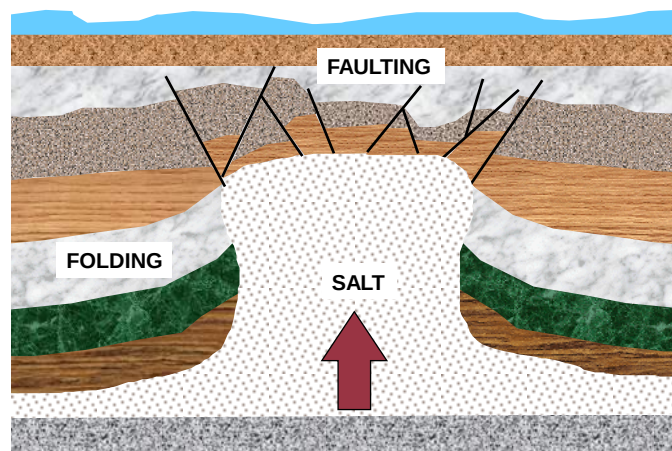


San Andreas Fault, USA

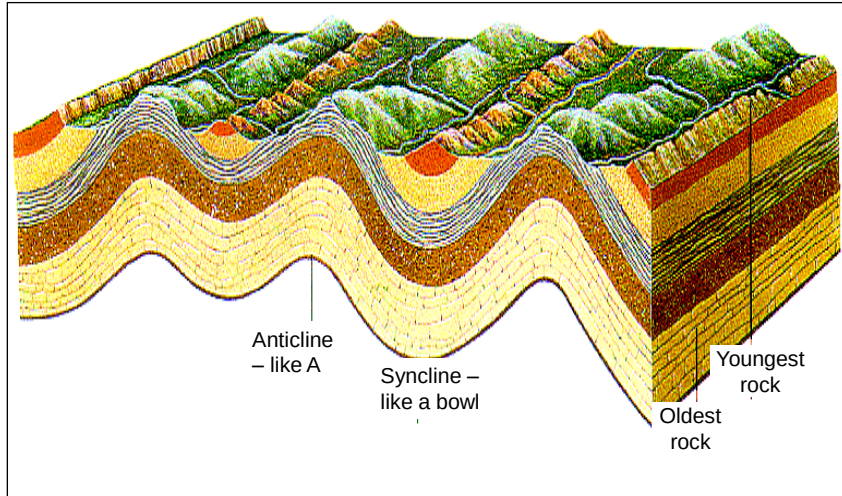
- Strike-Slip fault
- Sliding plate boundary



Salt Diapirism



Fold Terminology



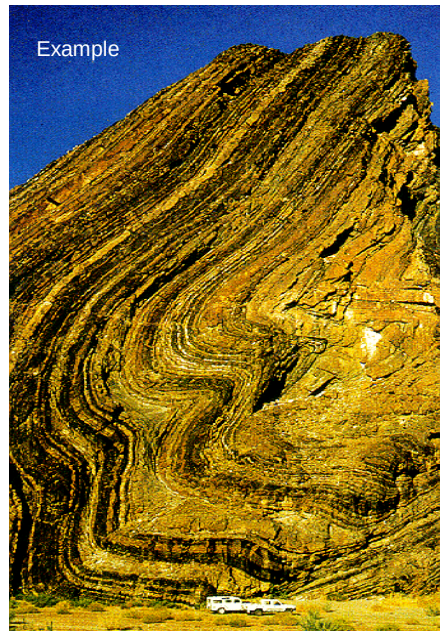
Anticline



Folding



Overtured Folds



- In a relatively relaxed geologic region, such as a young deltaic sedimentary basin, σ_1 and σ_2 tend to be approximately equal. If the sediments are assumed to behave elastically then
 - $\varepsilon_x = \sigma_x/E - \mu\sigma_y/E - \mu\sigma_z/E$
- For compressed rock due to sedimentation $\varepsilon_x = 0$
- Also, since $\sigma_x = \sigma_y$ then
 - $\sigma_x = \sigma_y = \mu/(1-\mu)\sigma_z$

Eaton Equation

- The fracturing pressure is the minimum pressure required to overcome the minimum principal stress.
- It is given as follows: $P_{ff} = \sigma_{min} + P_f$
- But $\sigma_{min} = \sigma_x = \mu/(1-\mu)\sigma_z$, and $\sigma_z = \sigma_{overburden} - P_f$
- Hence, $F = P_{ff}/D = \mu/(1-\mu)[\sigma_z - P_f]/D + P_f/D$

F = fracture gradient, P_f = pore pressure

D = depth